

02 Invisible Barriers

An Interactive Installation Translating Urban Accessibility Through Touch and Sound



2026.1-2026.9

This project investigates why accessibility infrastructure often fails in everyday urban environments and how these failures affect the independent mobility of visually impaired people.

Through public interviews, case studies and spatial research, I found that the problem extends beyond damaged, obstructed or poorly maintained tactile paving. It is also embedded in an urban system that frequently treats able-bodied and vision-based experience as the default. When accessible routes become fragmented or unreliable, seemingly minor spatial barriers accumulate into additional detours, dependence on others, psychological pressure and reduced mobility.

In response, I translated common urban obstacles, tactile paving patterns and public infrastructure into a modular tactile language. These elements were developed into an interactive wall installation combining touch and sound. Visitors explore different surfaces through their hands, triggering corresponding sound responses and experiencing the urban environment through a non-visual sensory system.

The project uses sensory translation to make overlooked accessibility issues more perceptible, asking: Who is excluded when cities are primarily designed around those who can see and move without barriers?

INSPIRATION

Damaged Tactile Paving and Inadequate Urban Maintenance

Broken, blocked, and poorly maintained tactile paving continues to affect the daily mobility of blind and visually impaired pedestrians. These cases reveal how inaccessible infrastructure, weak maintenance systems, and careless occupation of walking routes can turn a basic guidance path into a source of risk and exclusion.

News 1 : Parked vehicles and street barriers repeatedly block tactile paving routes.

News 2 : Cracked tiles, missing guide strips, and poor repairs create daily safety hazards.

News 3 : Better inspection, timely maintenance, and clear accountability are urgently needed.

Mobility Barriers for the Visually Impaired

1. Social Infrastructure

Tactile paving often ends abruptly, detours around obstacles, and fails to connect to entrances.

2. Maintenance Problems

Broken tiles, worn warning dots, and long-term occupation by vendors or shared bikes reduce accessibility.

3. Over-Reliance on Vision

Many intersections lack audible signals. Traffic noise hides direction cues, and digital navigation is not fully screen-reader accessible.

4. Cultural Background

Visual information is treated as the default. Independence is underestimated, job stereotypes persist, and public contact remains limited.

RESEARCH

Public Interview



I Never Noticed

I used to walk past tactile paving, without thinking about who relies on it. Only after hearing blind people's stories did I realize how many obstacles are invisible to me.

Hard to Imagine

Even when I try to understand, I still cannot truly feel the stress of moving through streets, crossings, and crowds without sight. It is difficult to empathize with an experience my body has never lived.



Too Easy to Assume

Many of us think offering help is enough, but we rarely notice how poor design makes daily travel exhausting. From the outside, it is easy to underestimate what blind people face every day.

Accessibility barriers are often overlooked because sighted people rarely experience the everyday challenges faced by visually impaired individuals.

01 What obstacles appear on tactile paving?

Common obstacles on tactile paving

Other obstacles

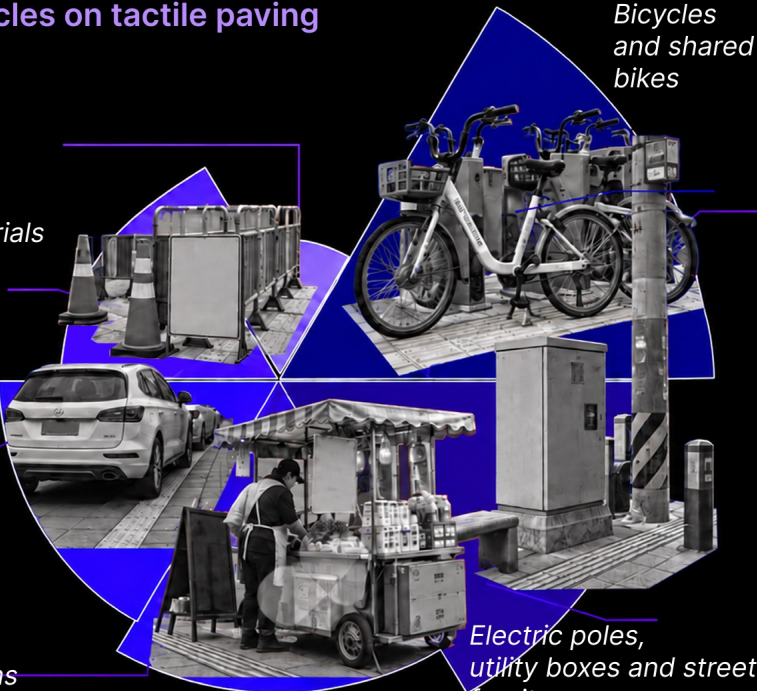
Construction materials and temporary facilities

Parked vehicles

Street vendors and storefront items

Bicycles and shared bikes

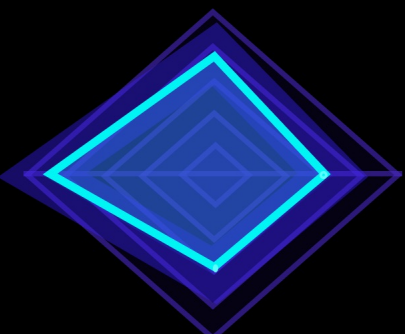
Electric poles, utility boxes and street furniture



03 Why is tactile paving occupied by facilities for sighted people?

Lack of Empathy

Car-Centric Planning



Accessibility as an Afterthought

Tactile paving is widely obstructed because accessibility is often overlooked in everyday urban planning.

WHY IS THE PHYSICAL ENVIRONMENT GETTING WORSE FOR DISABLED PEOPLE?

SOCIOCULTURAL MECHANISMS

ABLE BODIES AS THE DEFAULT

Design overlooks diverse bodies, leaving many barriers ignored or superficially addressed.

INVISIBLE NEEDS CYCLE

Barriers reduce outings, making real accessibility needs less visible.

ACCUMULATING BARRIERS

Unresolved barriers alter routes and choices, creating further physical, social, and psychological obstacles.

PSYCHOLOGICAL FACTORS OF DISABLED PEOPLE

ANXIETY DRIVES ROUTE AVOIDANCE

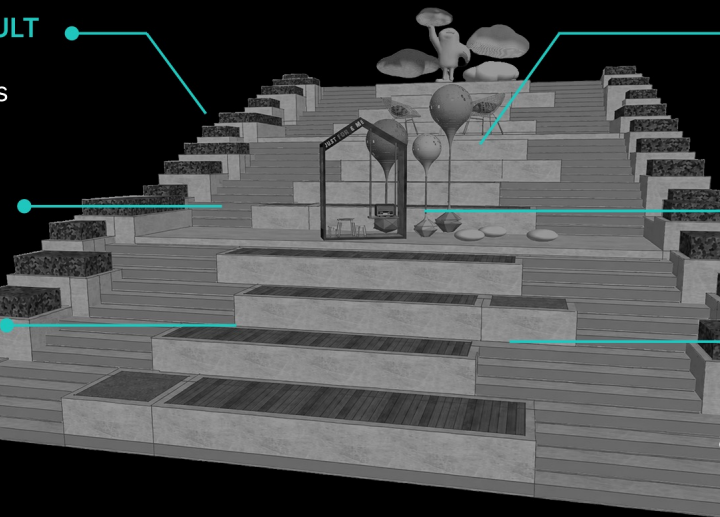
Uncertain routes and sudden barriers discourage independent travel.

GUILT SILENCES NEEDS

Fear of burdening others discourages complaints and feedback.

HELPLESSNESS CAUSES WITHDRAWAL

Repeated barriers reduce travel and social participation, making needs increasingly invisible.



Society treats able-bodied people as the default standard, making the spatial experiences of disabled people difficult to see. Facility gaps then force them to reduce their outings, making their needs even less visible.

CONSEQUENCES OF INVISIBLE BARRIERS

1

Route Disruption

Missing or unusable facilities suddenly interrupt planned routes.

2

Detours Or Assistance

Visually Impaired users must reroute or depend on others.

3

Repeated Barriers

Detours often lead to more steps, narrow passages, or unusable facilities.

4

Reduced Travel

Repeated barriers increase costs and dependence, discouraging independent outings.

One unresolved barrier creates a cycle of physical, behavioural, social, and psychological obstacles.

BARRIERS PROLIFERATE LIKE ORGANISMS

01 | Trigger

One unresolved facility gap interrupts an accessible route, forcing detours, waiting, or dependence.

02 | Spread

Changed routes expose users to further steps, narrow passages, and unusable facilities. One barrier spreads into behavioural, relational, and psychological effects.

03 | Proliferation

When barriers remain unseen and unresolved, they accumulate, occupy more areas of life, and shrink independent mobility.

JOURNAL ARTICLE

Care Recipient Concerns About Being a Burden and Unmet Needs for Care

Anna Vazeou Nieuwenhuis, MS, Scott R Beach, PhD, Richard Schulz, PhD

Innovation in Aging, Volume 2, Issue 3, September 2018, igy026, <https://doi.org/10.1093/geroni/igy026>

Results: Findings indicated a substantial number of inaccessible elements for MobAD users in public spaces. Pathway characteristics, boarding ramps, entrance features, confined spaces, and service surfaces were deemed to be the least accessible elements. These barriers had multifaceted effects on MobAD users' quality of life with aspects of physical health, mobility, and use of public transport being most affected.

Results: The following themes emerged: reduced physical burden on family/friends and increased independence and freedom. However, an EPIOC does not eliminate other practical problems, particularly during transportation and negotiating kerbs and slopes. Users also reported anxiety, in relationship to EPIOC use, e.g., weather conditions, personal safety (muggings), use of ramps and kerbs.

27.2%

43.6%

Somewhat Concerned

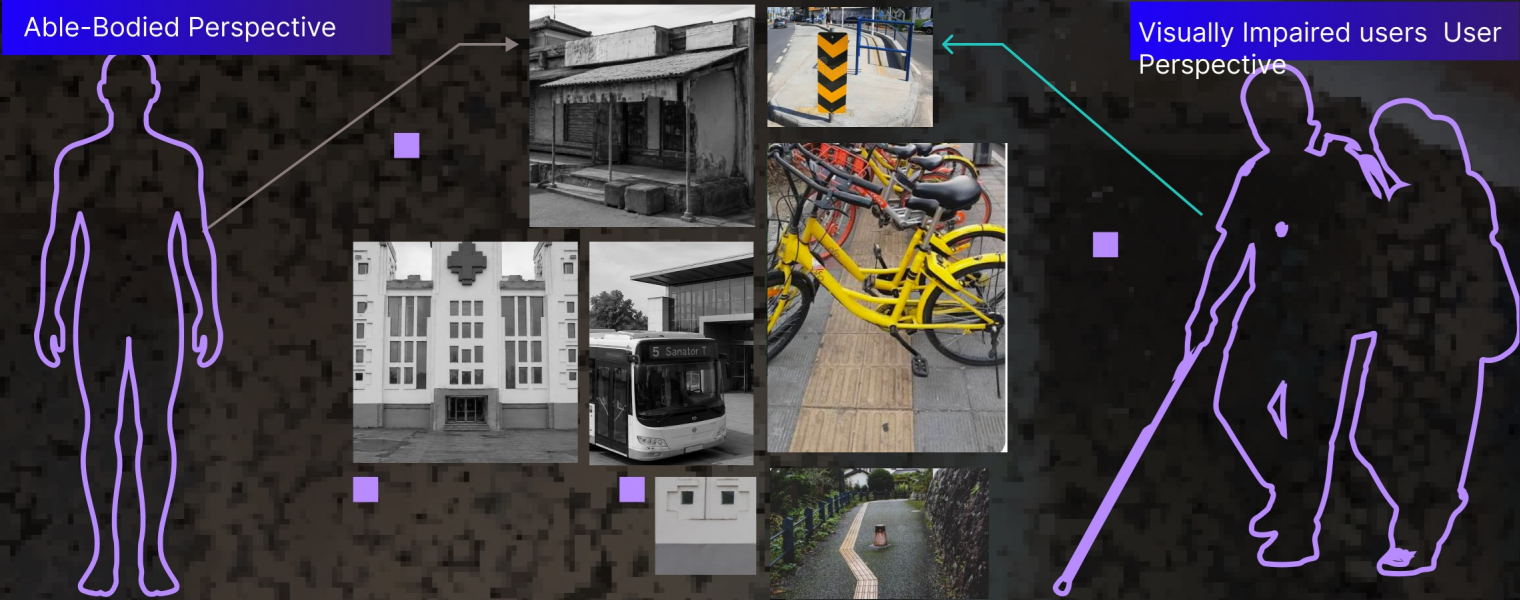
Very Concerned

Fear of Burdening Others



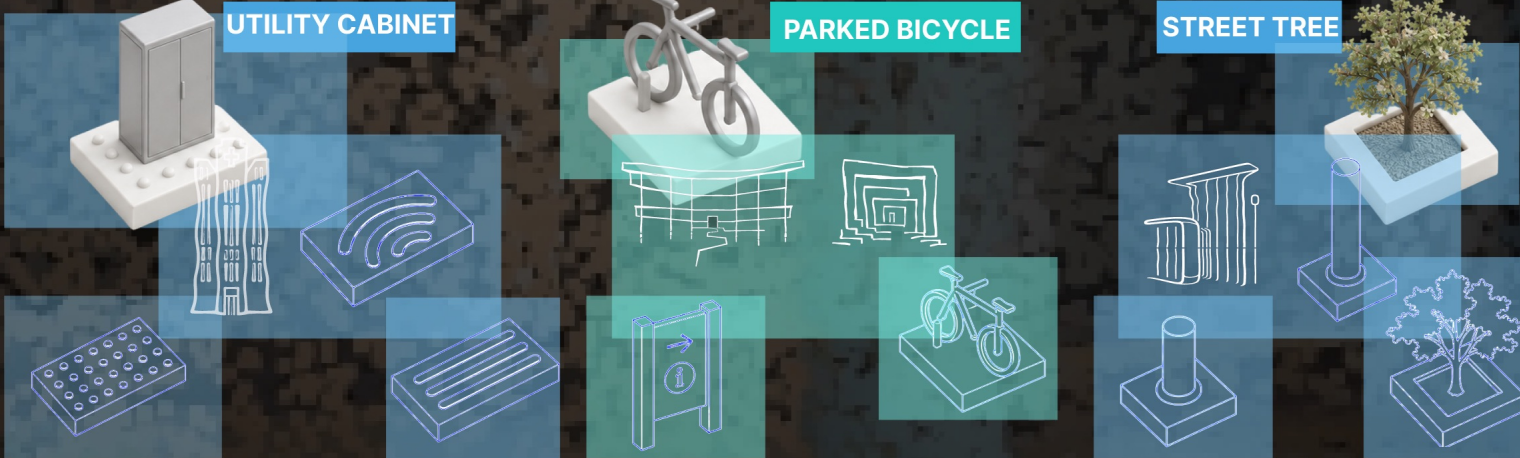
IDEATION

Spaces are often designed around able-bodied users, meaning that many barriers only become apparent when people with different bodily needs try to navigate them. My design translates these differences in embodied experience into something the audience can experience firsthand.

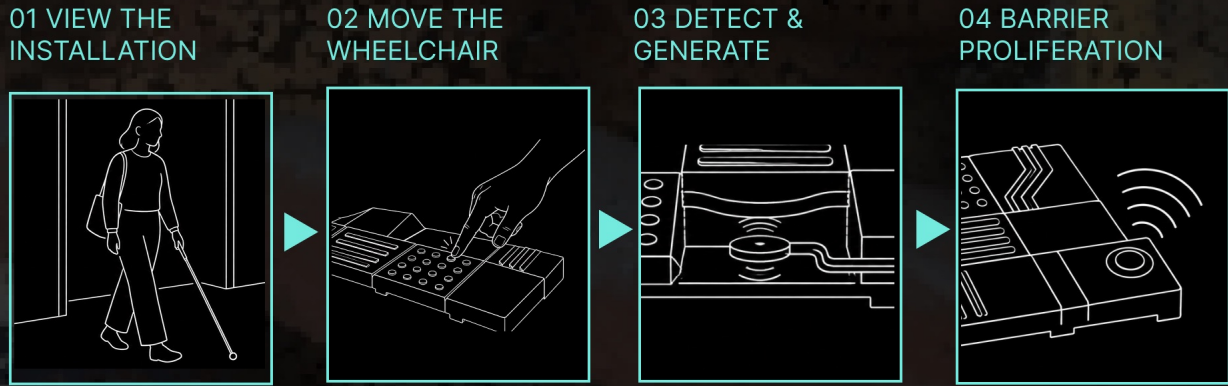


CONCEPT DEVELOPMENT

In response, I translated common urban obstacles, tactile paving patterns and public infrastructure into a modular tactile language.

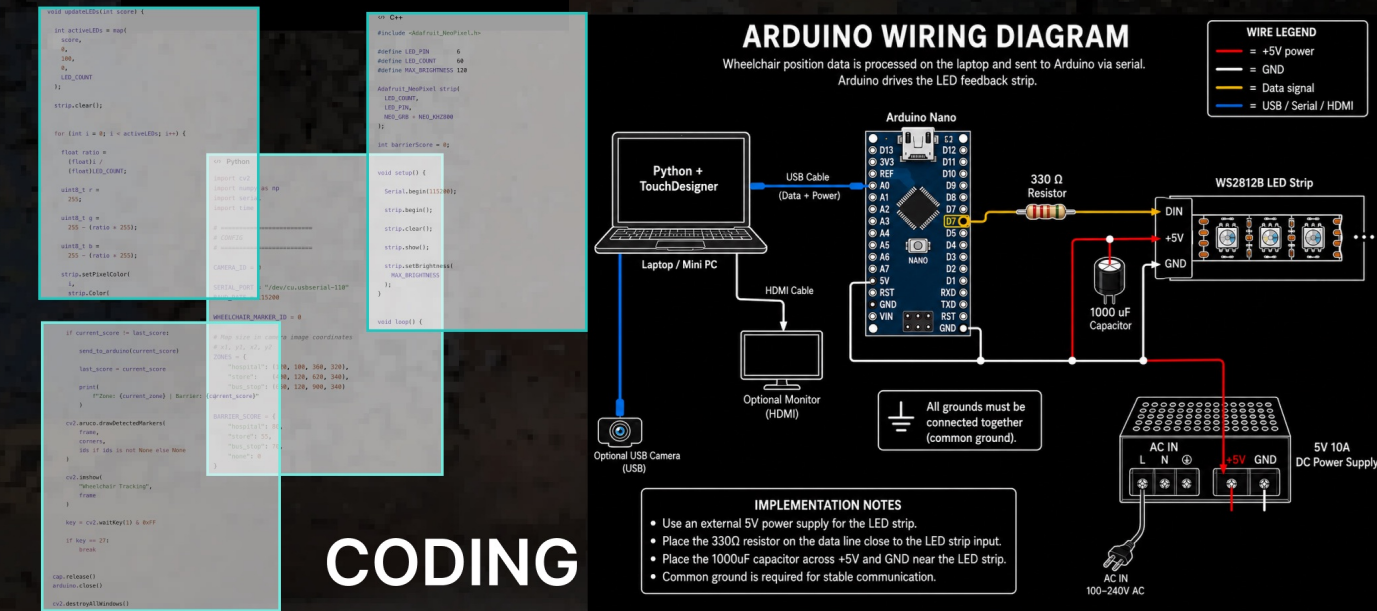
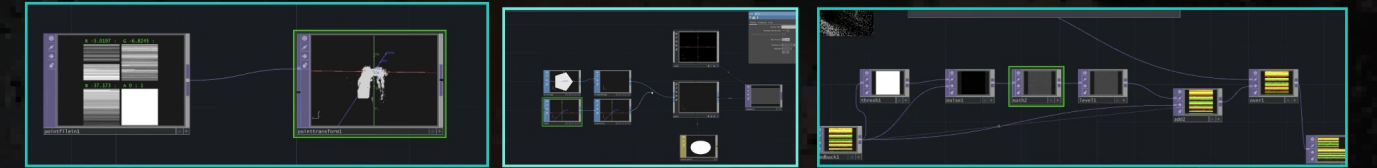


IINTERACTIVE LOGIC



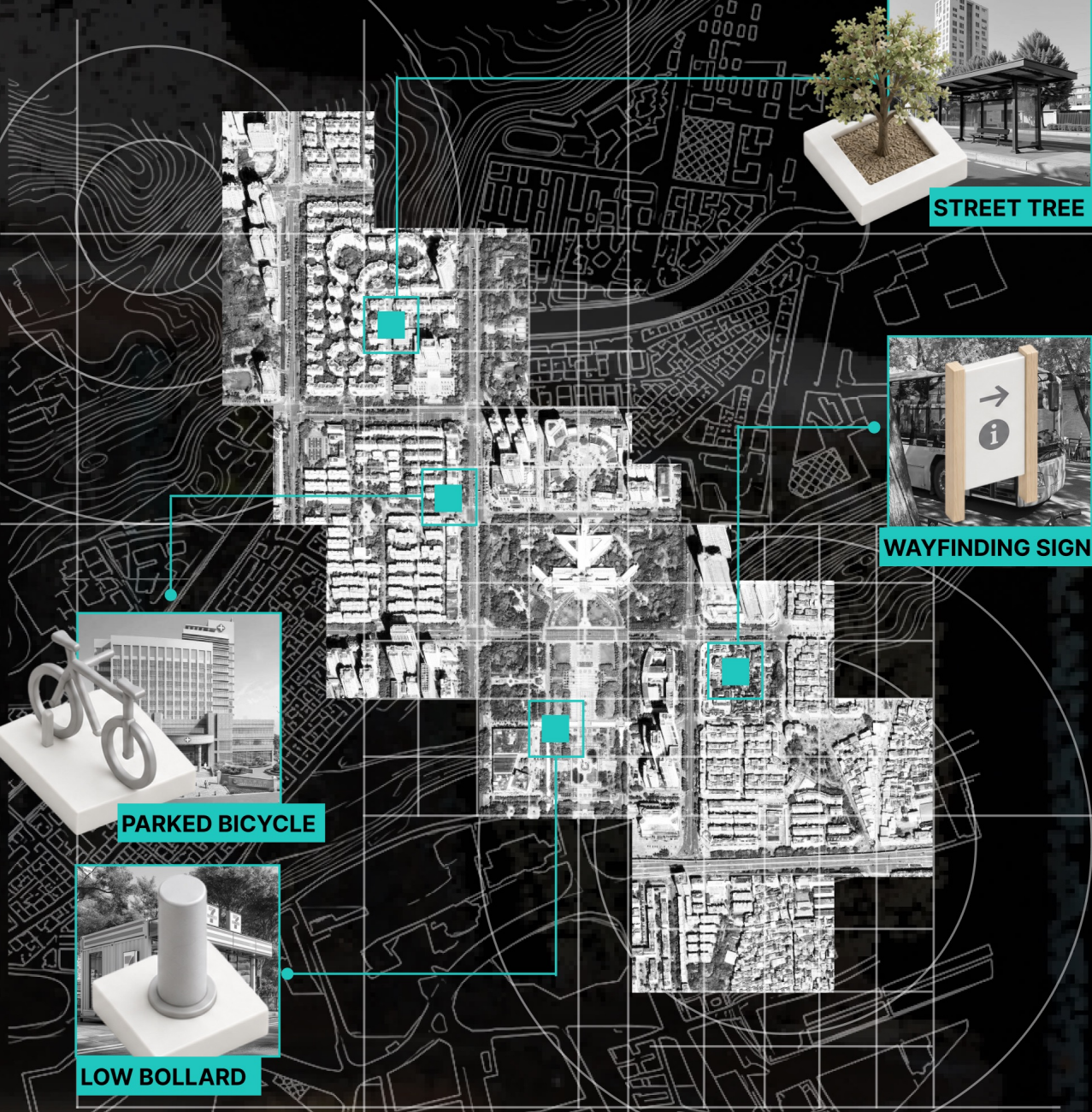
TECHNICAL SUPPORT

The technical system combines TouchDesigner, computer vision, and Arduino to track hands movement and translate spatial barriers into real-time visual and physical feedback.



CODING

MAP DESIGN



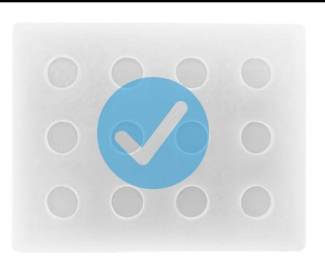
3D DESIGN

In response, I translated common urban obstacles, tactile paving patterns and public infrastructure into a modular tactile language.



Material Experiment Design in Blender

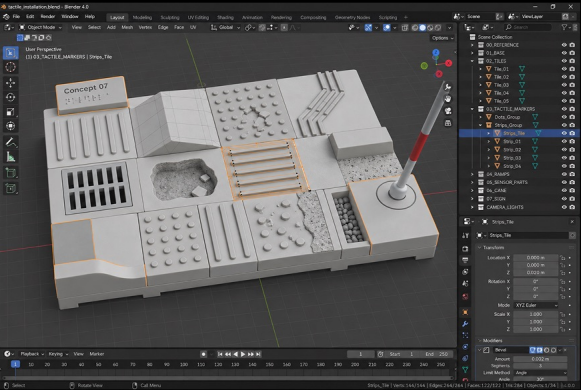
Material Feasibility Validation



Soft, providing a gentle and comfortable tactile experience.



Too cold and rigid, making it unsuitable for prolonged touch.



Who is excluded when cities are primarily designed around those who can see and move without barriers?

MODULE DESIGN EXPLORATION

Based on the urban barriers identified through research, I extracted and simplified tactile paving patterns and common street objects into a modular visual language. These elements were tested through different forms, scales and combinations, forming the basis of the final interactive installation.

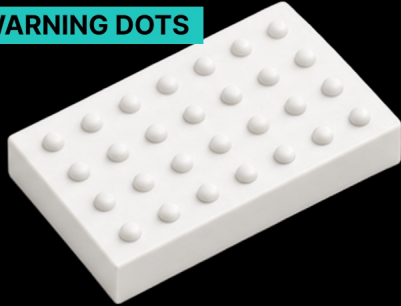
LINEAR GUIDANCE



CURVED GUIDANCE



WARNING DOTS



STREET TREE



TALL BOLLARD



LOW BOLLARD



PARKED BICYCLE



WAYFINDING SIGN



UTILITY CABINET



Sound Design : Artist Research



Music Album



Audio Playback System



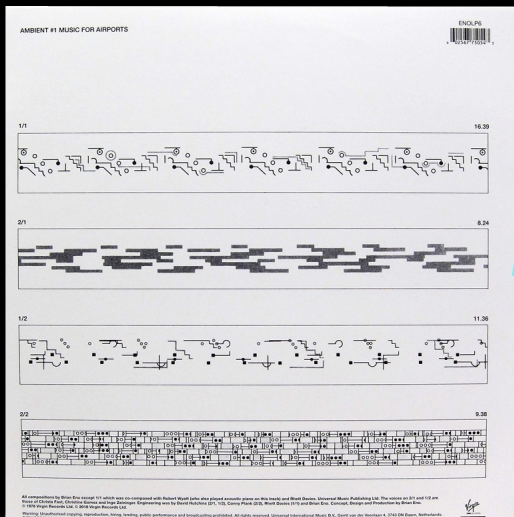
Musical System Diagram

Artist name : Brian Eno

Brian Eno's practice shifts music from fixed compositions to system-based generation, where sound emerges through looping structures and temporal relationships. Instead of focusing on melody and rhythm, his work emphasizes texture, duration, and spatial atmosphere, turning music into an environmental condition.

CASE STUDY

This diagram does not represent musical notation, but a system of temporal relationships between sound events.



Composing sounds through modular graphic systems

Sound Events (Trigger)



Timbre Layer (Texture)



Sound Duration / Sustain



Different Loop Systems



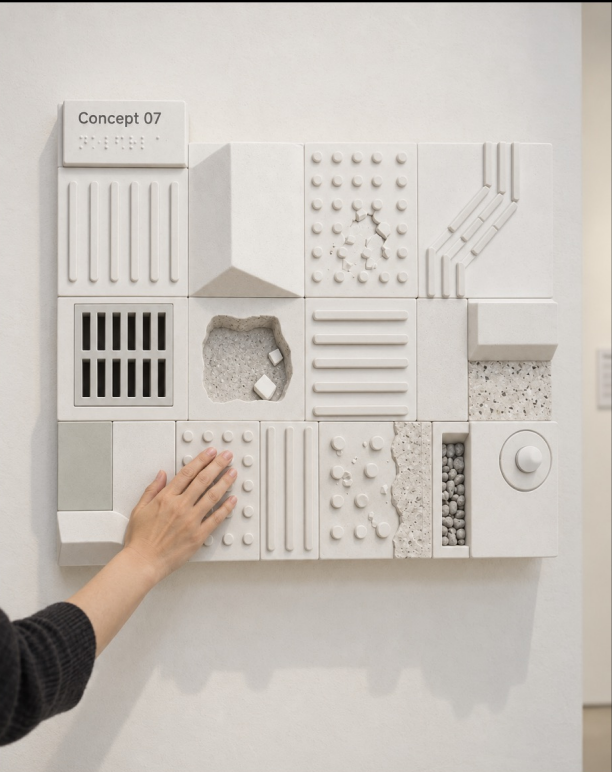
FINAL DISPLAY



When accessible routes become fragmented or unreliable, seemingly minor spatial barriers accumulate into additional detours, dependence on others, psychological pressure and reduced mobility.



In response, I translated common urban obstacles, tactile paving patterns and public infrastructure into a modular tactile language.



Who is excluded when cities are primarily designed around those who can see and move without barriers?

FEEDBACK



"I usually notice tactile paving, but I never really think about what happens when the route is suddenly interrupted. Touching the different surfaces made me realise how small physical changes can completely affect the way someone navigates a space."

"The combination of touch and sound made the experience more immersive, but I wanted the relationship between each texture and sound to be clearer. A stronger connection could help me understand what each module represents in the real urban environment."



In future development, I would involve visually impaired participants more directly through co-design and iterative testing, refine the sound–texture mapping, and explore how the modular system could represent a wider range of urban accessibility barriers.

REFLECTIVE CONCLUSION

This project began with the visible problem of damaged and occupied tactile paving, but gradually developed into a broader investigation of how accessibility can become invisible within everyday urban systems. Through the design process, I realised that the aim should not be to simulate the experience of blindness, as a temporary interaction can never reproduce the complexity of lived experience. Instead, the installation works as a sensory translation, using touch and sound to make overlooked spatial barriers more perceptible to sighted audiences. Audience feedback also revealed that interaction alone is not enough. The relationship between each tactile module, its sound response and the real-world barrier it represents needs to be more explicit. In future development, I would involve visually impaired participants more directly through co-design and iterative testing, refine the sound–texture mapping, and explore how the modular system could represent a wider range of urban accessibility barriers.